

**Department of Chemical and Biomolecular Engineering
North Carolina State University**

CHE 330

Evaluation of Fluidized Bed Characteristics

1. Introduction

A fluidized bed is a mass of solid particles suspended by the upward movement of a fluid (often a gas). The suspended fluidized particles collectively behave as a liquid, significantly changing the transport phenomena and thermodynamic properties of the material. Mass and heat transfer are rapid due to the vigorous mixing of particles, allowing for nearly uniform temperature and concentration distributions throughout the bed. These characteristics make fluidized beds ideal for use in many areas of engineering, including petroleum refining, chemical synthesis and combustion.

2. Objective

The objective of this experiment is to determine fluidization characteristics and heat transfer properties for a fluidized bed of Al_2O_3 particles.

3. Theory

An overview of a fluidized bed is shown in Figure 1. The solid particles in the bed are pulled toward the ground by the force of gravity. The terminal velocity (u_t), i.e. the maximum velocity able to be achieved by a solid particle due to gravity, is given by

$$u_t = \frac{(\rho_s - \rho_g)gd_p^2}{18\mu} \quad (1)$$

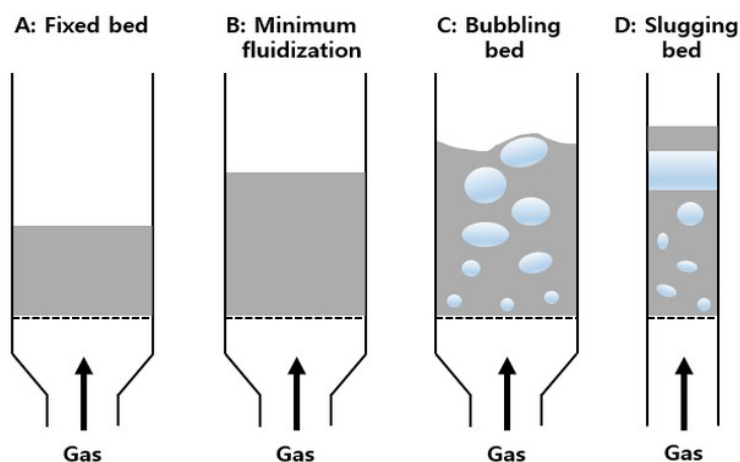


Figure 1. Overview of fluidized bed operation (adapted from [1]).

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where ρ_s is bulk solid density, ρ_g is gas density, g is the gravitational acceleration constant (9.8 m/s²), d_p is particle diameter and μ is gas viscosity.

In contrast to the force of gravity, gas flow entering from the bottom of the bed imparts an upward force to the particles. Once the upward pressure force of the gas flow on the particles is greater than the downward pull of gravity, the particles will begin to rise and float. This force balance is described by [2]

$$\Delta P = (\rho_s - \rho_g)(1 - \varepsilon)gL \quad (2)$$

where ΔP is pressure drop across the particle bed, ε is the void fraction of the particle bed, and L is the height of the bed.

Pressure drop across a packed bed can be estimated by the Ergun equation [2]:

$$\frac{\Delta P}{L} = \frac{150\mu}{d_p^2} \frac{(1 - \varepsilon)^2}{\varepsilon^3} u_0 - \frac{1.75\rho_g}{d_p} \frac{(1 - \varepsilon)}{\varepsilon^3} u_0^2 \quad (3)$$

The gas velocity (typically referred to as superficial gas velocity, u_0) where upward force on the particles overcomes downward force is known as the minimum fluidization velocity (u_{mf}) which can be calculated theoretically by combining Equations 2 and 3:

$$g(\rho_s - \rho_g) = \frac{150\mu(1 - \varepsilon_{mf})}{d_p^2 \varepsilon_{mf}^3} u_{mf} + \frac{1.75\rho_g}{d_p \varepsilon_{mf}^3} u_{mf}^2 \quad (4)$$

where ε_{mf} , the void fraction of the particle bed at minimum fluidization, is given by [3]

$$\varepsilon_{mf} = 1 - \frac{\rho_s}{\rho_p} \quad (5)$$

where ρ_p is the particle density. Note that for most solid particles ρ_p is larger than ρ_s since ρ_s measures the bulk density of the solid (solid + voids) while ρ_p is the density of only the particle itself.

The value of u_{mf} for a bed of particles can also be determined empirically by examining pressure drop behavior of the bed as a function of u_0 ; this plot of u_0 vs. ΔP is known as a “fluidization curve.” Fluidization curves tend to have the shape found in Figure 2. The u_0 value where pressure drop becomes fairly constant is the empirical u_{mf} . Empirical u_{mf} values often vary significantly from theoretical values due to non-ideal mixing, especially for small laboratory-scale fluidized beds.

At superficial gas velocities above u_{mf} , the particle bed becomes fluidized and the gas-solid mixture begins to behave like a liquid. As u_0 increases, the behavior of the fluidized bed changes,

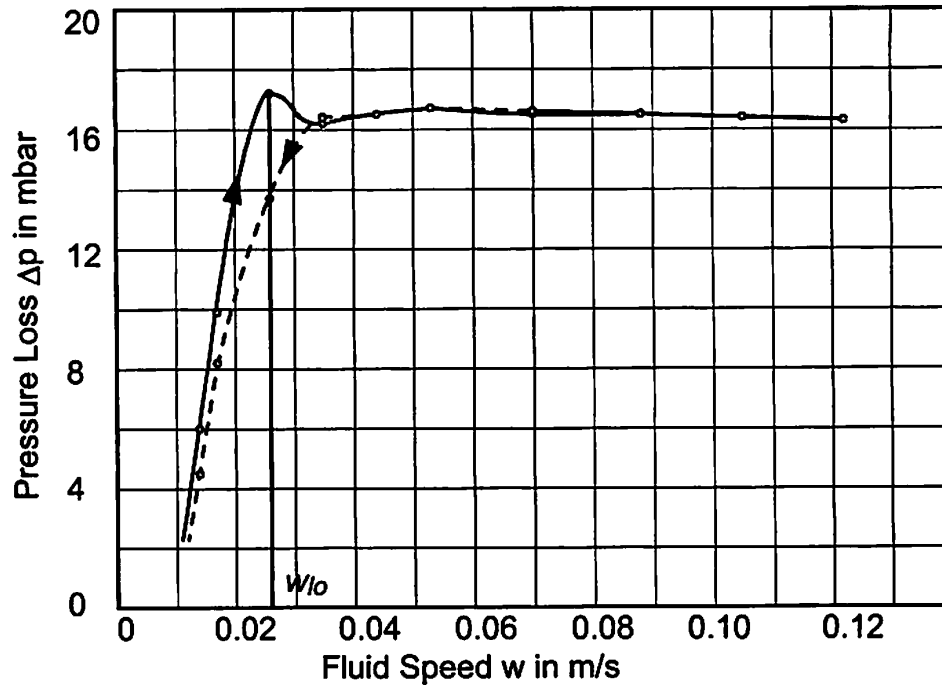


Figure 2. Typical fluidization curve indicating an approximate u_{mf} of 0.025 – 0.035 m/s. The solid line indicates data taken with increasing fluid speed, while the dashed line indicates data taken with decreasing fluid speed [3].

proceeding through different fluidization regimes as indicated in Figure 1. A brief description of each regime is given in order of increasing u_0 :

- 1) Packed bed regime (u_0 below u_{mf}) – no particle fluidization
- 2) Smooth regime – small bubbles are seen exiting the surface of the bed, but in general the particles are not vigorously mixed. Operation in the smooth regime is ideal for powder-coating applications [4].
- 3) Bubbling regime – numerous bubbles of significant size can be seen emerging from the bed and the suspended particles are vigorously mixed. Mass and heat transfer rates within the bed are higher than for the smooth regime. Operation in the bubbling regime is typical for many high-temperature exothermic reactions using heterogeneous catalysts.
- 4) Slugging regime – very large bubbles causing significant voids within the bed begin to emerge from the surface.
- 5) Pneumatic transport – one very large bubble is formed which displaces the entire bed of particles and conveys the bed downstream.

Fluidization characteristics for a given bed of particles also vary depending on bulk properties of the particle, which plays a crucial role in determining the feasibility of application of fluidized beds in industry. Particles are classified into four different groups (known as “Geldart groups”), as described in Figure 3 [4]:

- Group A – Non-agglomerating low-density particles between 30 and 100 μm . Particles readily aerate and fluidize easily (example: FCC catalyst).

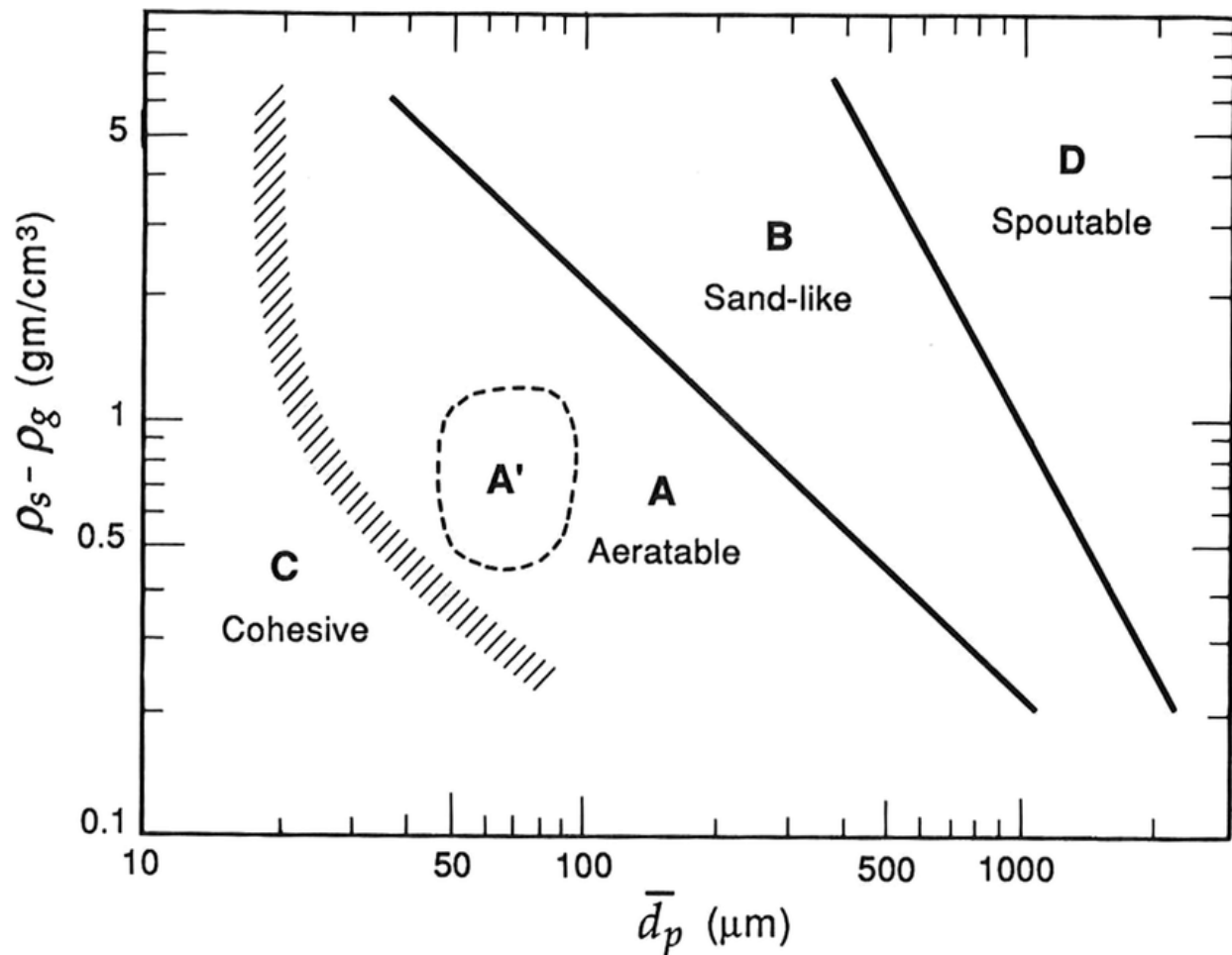


Figure 3. Geldart classification of particles in ambient air [4]. Region A' is a range of properties for well-behaved Fluidized Catalytic Cracking (FCC) catalyst. [4]

- Group B – Rigid particles between 100 and 1000 μm . Ideal for fluidization, bubbles form quickly to produce vigorous bubbling under proper conditions (example: sand).
- Group C – Cohesive low-density particles between 0 and 30 μm . Interparticle forces exceed that from gravity or rising gas, producing clumping during fluidization (example: flour).
- Group D – Dense particles above 1000 μm which “spout” rather than achieving ideal fluidization (example: coffee beans).

Generally, only Group A and B powders are considered for fluidized beds in industrial applications.

Fluidization characteristics impact rates of mass and heat transfer within the bed. In particular, heat transfer properties of the fluidized bed vary significantly due to factors including:

- fluid and particle properties
- superficial gas velocity
- Geldart particle classification

The coefficient of heat transfer (α) for the fluidized bed can be calculated from empirical observations by Equation 6:

$$\alpha_{\text{exp}} = \frac{Q}{A_h \Delta T} \quad (6)$$

where α_{exp} is the empirical fluid bed heat transfer coefficient, Q is heater power, A_h is the surface area of the heater, and ΔT is represented by

$$\Delta T = T_h - T_{\text{bed}} \quad (7)$$

where T_h is the heater temperature and T_{bed} is the fluid bed temperature [4].

4. Experimental

4.1. Equipment

The fluidized bed apparatus used in experimentation is shown in Figure 4. A glass cylinder (labeled “1”) with a 95 cm² cross-section contains a bed of α -Al₂O₃ particles (labeled “2”, $d_p = 100 \mu\text{m}$, $\rho_p = 4000 \text{ kg/m}^3$, $\rho_s = 1760 \text{ kg/m}^3$). Compressed air enters the unit from a house air line regulated to a pressure of approximately 6 bar. A throttling valve (V2) is used to control inlet pressure, which is monitored using a pressure gauge (P3). Air flow rate is controlled by a needle valve (V1) and measured using a rotameter (F1) before reaching a gas distributor chamber (labeled “3”) below the bed. Pressure in the gas distributor chamber is measured using a pressure gauge (P2). A rupture disk (V3) is used as a safety device to limit pressure in the gas distributor to 0.5 bar.

Above the gas distributor chamber is a porous sintered metal frit used to distribute gas evenly across the bottom of the bed. The rising air fluidizes the α -Al₂O₃ bed and exits the cylinder through filters (labeled “4”) to prevent particle carryover. An adjustable 100 W heated cylinder (labeled “5,” $A_h = 13.35 \text{ cm}^2$) immersed in the particles is used to provide heat to the surrounding bed. A variac (E1) is used to regulate heater power. Heater, bed and inlet air temperatures are measured by thermocouples (T1, T2 and T3, respectively).

4.2. Experimental procedure

4.2.1. Fluidization study

- 1) Supply power by turning the red dial on the left-hand side of the experimental unit.
- 2) Verify that the glass cylinder is filled with α -Al₂O₃ particles to a bed height of approximately 9 cm.
- 3) Verify that the air regulator (located behind and above the experimental unit) is adjusted to a delivery pressure of approximately 6 bar.
- 4) Adjust the inlet pressure (P3) to 1 bar by adjust pressure regulator V2.

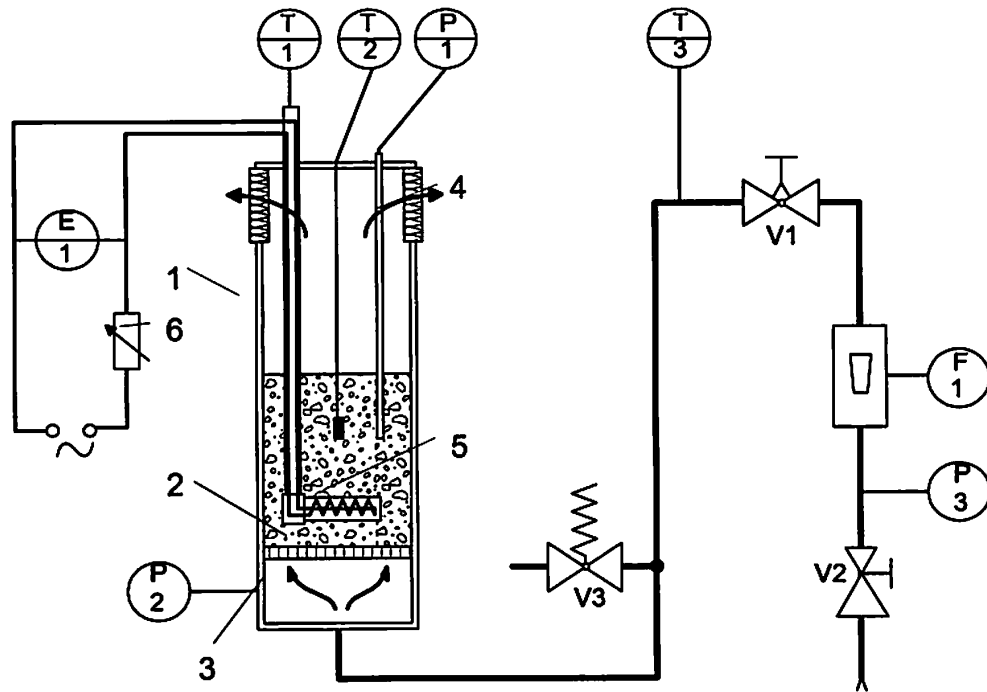


Figure 4. Schematic of experimental fluidized bed apparatus [1].

- 5) Using the flow regulating valve V1, adjust the air flowrate to the minimum reference value of 8 L/min (labeled “Ref” on the rotameter). Read the rotameter using the bottom of the float as the indicator.
- 6) Once the fluid bed pressure reading (P1) is stable, record the pressure, flowrate and bed height.
- 7) Increase the flowrate to 9 L/min by adjusting V1, then adjust inlet pressure P3 to 1 bar using V2 if necessary. Once the fluid bed pressure reading (P1) is stable, record the pressure, flowrate and bed height.
- 8) Continue the procedure of increasing flowrate in small (~ 1 L/min) steps while recording pressure, flowrate and bed height and observing the bed. After each adjustment, be sure to adjust inlet pressure P3 to 1 bar using pressure regulator V2.
- 9) As soon as particles on the surface of the bed begin to move (i.e., a bubble escapes or a crack is visible in the bed), monitor the pressure P1. If the pressure stays constant or begins to fall, it is likely that the bed loosening velocity (u_{loose}) has been reached. Record the bed loosening velocity observed during testing. *Note: u_{loose} is defined as the velocity at which particles begin to fluidize, and is a rough estimation of u_{mf} based solely on visual observations of particle behavior within the bed while increasing the air flowrate.*
- 10) Continue the procedure of increasing flowrate in small (~ 1 L/min) steps while recording pressure, flowrate and bed height until a flowrate of 20 L/min is reached. If any of these values “jump” continuously (they likely will), record a mean value. Observe the bed behavior, taking note of the various fluidization regimes achieved during testing.
- 11) Once 20 L/min is reached, continue increasing flowrate in steps, but increase the step size to 5 L/min (e.g. 20 L/min, 25 L/min, 30 L/min, etc.). Record pressure, flowrate and bed

height until a flowrate of 70 L/min is reached. Observe the bed behavior, taking note of the various fluidization regimes achieved during testing.

- 12) Next, begin reducing the flowrate in the same steps (e.g. go from 70 to 20 L/min in 5 L/min steps, then from 20 to 8 L/min in 1 L/min steps) while recording pressure, flowrate and bed height once each reading has stabilized.

4.2.2. Heat transfer study

- 1) Verify that the black mark on the heating rod is at a height of approximately 18 cm.
- 2) Adjust air flowrate to 70 L/min using V1 and heater power to 30 W using the variac (E1).
- 3) Allow for temperatures to stabilize before recording heater temperature (T1) and fluid bed temperature (T2); this will likely take 3-5 minutes.
- 4) Decrease the air flowrate to 65 L/min using V1, adjusting pressure regulator V2 to keep inlet pressure P3 at a constant value of 1 bar. Maintain heater power at 30 W by adjusting variac E1. Allow for temperatures to stabilize before recording heater temperature (T1) and fluid bed temperature (T2).
- 5) Repeat Step 4 for air flowrates of 60, 55, 50, 45, 40, 35, 30, 25 and 20 L/min, allowing ample time for steady state to be reached before recording heater temperature (T1) and fluid bed temperature (T2). *Note: As the air flowrate is decreased, the heater temperature will increase. The heater is equipped with a governor that will temporarily cut off power to the heater if the heater temperature exceeds ~85-90°C. As a result, it is not possible to test air flowrates below 20 L/min during the heat transfer study.*

5. Data Analysis

Calculate theoretical u_t and u_{mf} values for the tested $\alpha\text{-Al}_2\text{O}_3$ particle. Produce a fluidization curve showing empirical ΔP and bed height on y-axes and superficial gas velocities on the x-axis. Determine an experimental u_{mf} value. Are the theoretical and empirical u_{mf} values similar? How does u_{mf} compare with u_{loose} ? Why does the fluidization curve look different depending on whether superficial gas velocities were raised from low to high values or lowered from high to low values during testing?

Identify each of the fluidization regimes achieved by the bed during testing. Specify the range of superficial gas velocities at which each regime exists as a multiple of u_{mf} (e.g. “the bubbling regime exists at superficial gas velocities between 5 and 7 times u_{mf} ”).

Hypothesize the Geldart group of the tested $\alpha\text{-Al}_2\text{O}_3$ powder. What experimental observations and material specifications lead you to this hypothesis?

Determine the empirical heat transfer coefficients for the bed as a function of u_0 . How do different fluidization conditions affect heat transfer within the bed?

References

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